

K-Nearest Neighbors

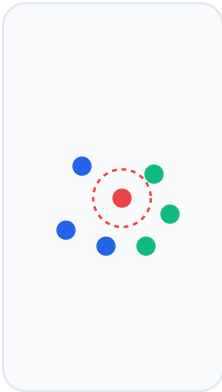
The Lazy Algorithm That Actually Works

Day 15 of 100 Days of Machine Learning

What is K-Nearest Neighbors?

KNN is a supervised algorithm for classification and regression.

Instead of learning a complex model, it predicts by finding the **K closest data points** to a new sample.



How Distance Works



KNN measures similarity usually via **Euclidean Distance**.

The closest neighbors have the greatest influence on the final prediction.

Impact of K Value

Small K (e.g. K=1)



- Sensitive to noise
- High Variance
- **Can overfit**

Optimal K



- Balanced boundary
- Best tradeoff
- **Good generalization**

Large K



- Highly smoothed
- High Bias
- **Can underfit**

Curse of Dimensionality

As the number of features (dimensions) increases, the distance between data points becomes almost equivalent. KNN loses its effectiveness in high-dimensional spaces.



Where is KNN Used?

- Recommendation Systems
- Image Classification
- Pattern Recognition
- Medical Diagnosis
- Fraud Detection
- Customer Segmentation

Python Implementation

```
from sklearn.neighbors import KNeighborsClassifier

# 1. Create
model = KNeighborsClassifier(n_neighbors=5)

# 2. Train
model.fit(X_train, y_train)

# 3. Predict
predictions = model.predict(X_test)
```

Advantages vs Disadvantages

- Advantages** ✓

 - ✓ Easy to understand
 - ✓ No explicit training phase
 - ✓ Works well on small datasets
 - ✓ Good for nonlinear boundaries
- Disadvantages** ✗

 - ✗ Slow predictions (large datasets)
 - ✗ Sensitive to feature scaling
 - ✗ Affected by noisy data
 - ✗ Poor high-dimensional performance



Job Seeker

- Know why **feature scaling** is essential.
- Understand how to choose the optimal **K value**.
- Classification (Majority) vs
- Curse of dimensionality impact on